# Task1 – Design and train image classification network

## 1.1 Dataset Description and Visualization

The Vehicle Classification dataset used in this task is a subset of the TAU Vehicle Type Recognition Competition, containing six common road vehicle categories: Bicycle, Car, Taxi, Bus, Truck, and Van. The training set contains 1909 images, while the validation set contains 180 images. Images are RGB with varying resolutions and are resized to 224×224 to match the ResNet50 input requirements.

## 1.2 Model Architecture

The classification network uses a ResNet50 backbone pre-trained on ImageNet as a feature extractor. The final fully connected layer is removed, replaced by:

1. **Global Average Pooling (GAP)**: Reduces the spatial feature map (2048, H/32, W/32) to a 2048-dimensional vector
2. **MLP Classifier Head**:
    - Linear layer (2048 → 1024)
    - ReLU activation
    - Dropout (p = 0.5) for regularization
    - Linear layer (1024 → 6)

The model contains **25,612,358** parameters. Table 1 summarizes the hyperparameters used for training.

Table 1: CNN Training Hyperparameters.

| Category | Parameter | Value / Range | Notes |
|---|---|---|---|
| **Loader Settings** | batch_size | 32 | Number of samples per batch |
| | num_workers | 0 | Number of data loader workers |
| **Data Augmentation** | hflip_prob | 0.5 | Horizontal flip probability |
| | rotation | 15 degrees | Max rotation angle |
| | color_jitter | [0.3, 0.3, 0.3, 0.1] | Brightness, contrast, saturation, hue |
| | translate | [0.1, 0.1] | Fractional translation (x, y) |
| | scale | [0.9, 1.1] | Scaling range |
| **Optimization** | lr | 1e-4 | Learning rate |
| | weight_decay | 1e-5 | L2 regularization |
| **Training** | num_epochs | 40 (Stopped at 32) | Total training epochs |
| | patience | 15 | Early stopping patience |

## 1.3 Training and Results

The model was trained for 32 epochs (early stopping triggered after 15 epochs of no improvement). The loss function used is Softmax Cross-Entropy Loss. Figure 1 shows the training curves over 32 epochs.

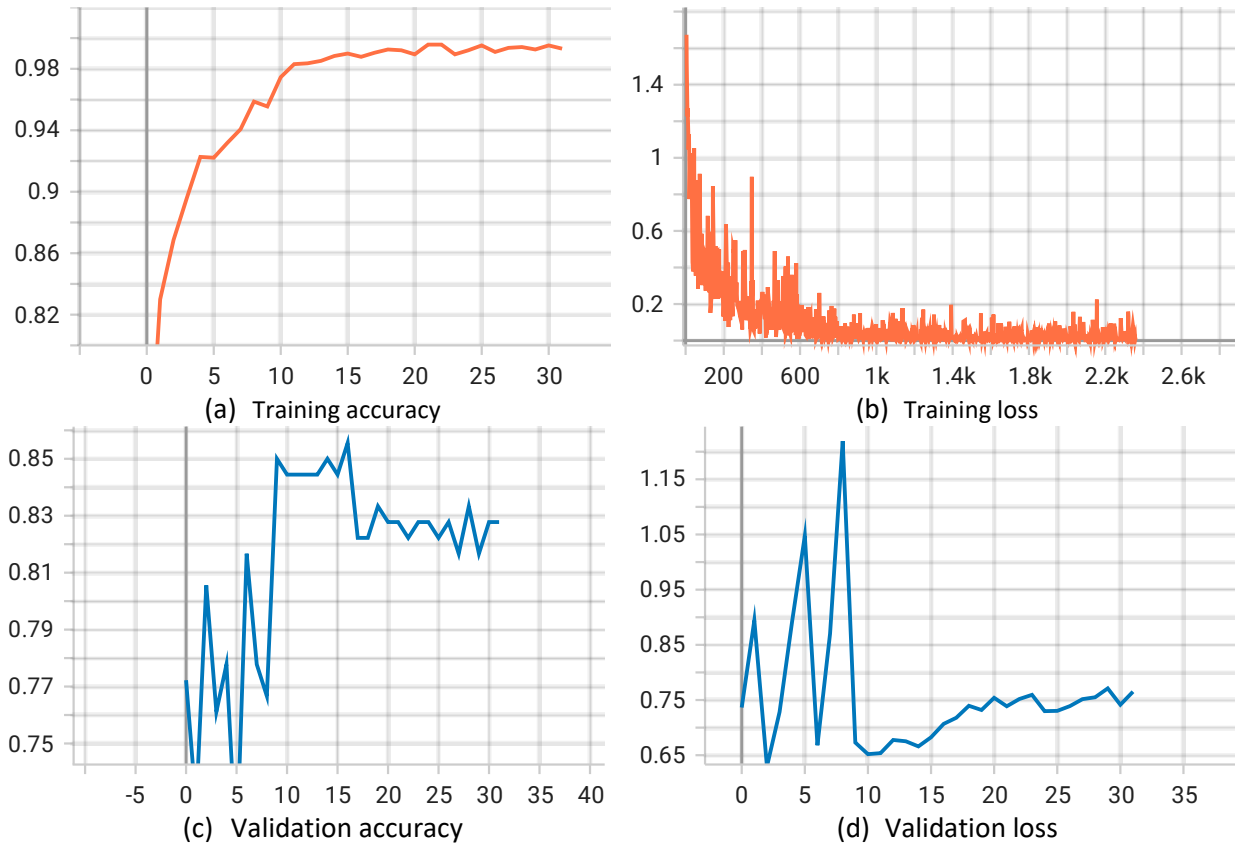


Figure 1: Training and validation loss/accuracy curves.

The model achieved strong performance on both training and validation sets, as shown in Table 2. Confusion matrix is also included in Figure 3 to better judge the model and see in which classes model underperformed.

Table 2: CNN Classification Performance.

| Single Task | Vehicle classification (Accuracy) |
|---|---|
| **Training** | 99.58 |
| **Validation** | 85.56 |
| **Download link** | cnn.pth |

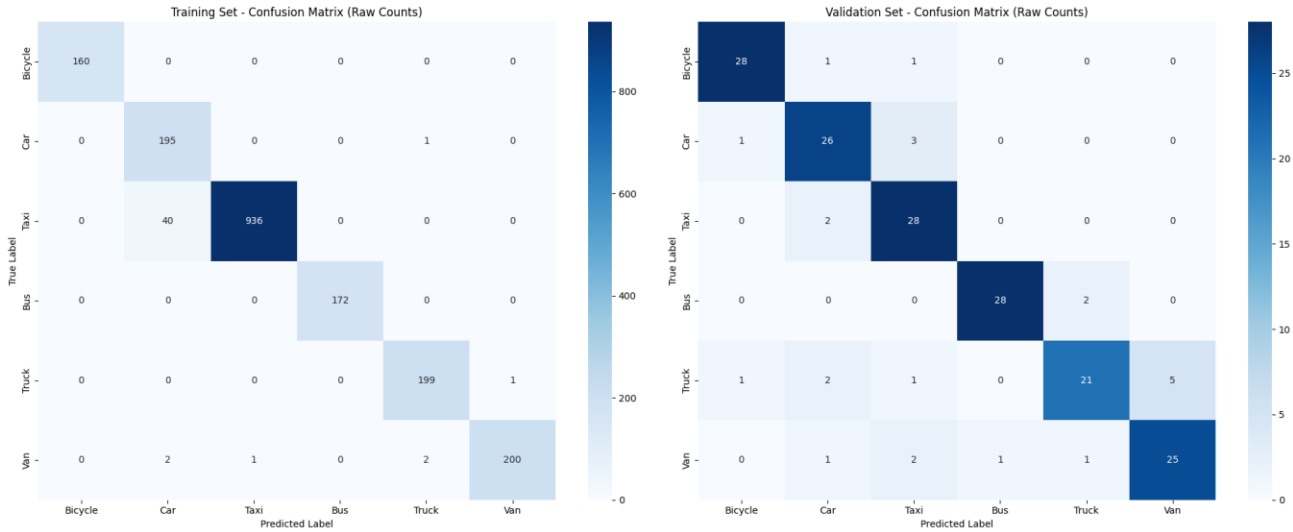


Figure 2: Confusion matrix.